MAST30027 Modern Applied Statistics – exam notes

9 Qs / 78 marks / 3 h ≈2.3 min/mark. Answers 2 dp, keep 4 dp inside. FX-82 has no Φ, χ^2, F tails: use printed **q*()** values.

1 • MLE → Fisher information → asymptotic distribution (Q1, 7–10 marks)

1. $\ell(\theta) = \sum_i \log f(x_i;\theta)$; drop θ-free terms ($+c$); collect $\sum x_i, \sum x_i^2, \sum \log x_i$. State support.
2. Score $\ell'(\theta) = 0 \Rightarrow \hat\theta$; check $\ell''(\hat\theta) < 0$.
3. $J(\theta) = -\ell''(\theta)$ (observed). $\mathcal{I}(\theta) = \mathbb{E}[J(\theta;X)]$, **whole sample** (contains n): replace $\sum X_i \to n\mathbb{E}X$, $\sum X_i^2 \to n\mathbb{E}X^2$. If ℓ'' has no data, write "$J = \mathcal{I}$, no expectation needed".
4. $\hat\theta \overset{\cdot}{\sim} N(\theta, \mathcal{I}(\theta)^{-1})$; est. var $1/\mathcal{I}(\hat\theta)$; Wald CI $\hat\theta \pm 1.96/\sqrt{\mathcal{I}(\hat\theta)}$.

Reparametrise: MLE invariant ($\widehat{g(\theta)} = g(\hat\theta)$); information not: redo, or $\mathcal{I}_\eta = \mathcal{I}_\theta (d\theta/d\eta)^2$. CI for $g(\theta)$: map endpoints (monotone g); better built on an unbounded scale (log, logit) then back-transformed. **Vector** θ: $\mathcal{I} = -\mathbb{E}[\partial^2\ell/\partial\theta\partial\theta^\top]$, $\mathrm{Var}\,\hat\theta \approx \mathcal{I}^{-1}$; $\left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-b^2}\left(\begin{smallmatrix} d & -b \\ -b & a \end{smallmatrix}\right)$. LR region $\{\theta : 2[\ell(\hat\theta) - \ell(\theta)] \le \chi^2_k(.95)\}$, $k = \dim\theta$.

n **iid model**	**MLE**	$\mathcal{I}$ **(whole sample)**
Bernoulli(p)	$\bar x$	$n/[p(1-p)]$
$Y_i \sim \mathrm{Bin}(m_i, p)$ same, in logit $\eta = \log\frac{p}{1-p}$	$\sum y_i/\sum m_i$ $\log\frac{\hat p}{1-\hat p}$	$\sum m_i/[p(1-p)]$ $\sum m_i\, p(1-p)$
Poisson(λ)	$\bar x$	n/λ
Exp(rate λ)	$1/\bar x$	n/λ^2
Exp(mean θ), $f = \theta^{-1}e^{-x/\theta}$	$\bar x$	n/θ^2
$N(\mu, \sigma^2)$, σ^2 known	$\bar x$	n/σ^2
$N(\mu_0, 1/\tau)$, μ_0 known, precision	$n/\sum(x_i-\mu_0)^2$	$n/(2\tau^2)$
$N(\mu_0, v)$, μ_0 known, variance	$\sum(x_i-\mu_0)^2/n$	$n/(2v^2)$
Gamma(shape a known, rate b)	$a/\bar x$	na/b^2
Geometric $p(1-p)^x$, $x \ge 0$	$1/(1+\bar x)$	$n/[p^2(1-p)]$
$\theta x^{\theta-1}$ on $(0,1)$	$-n/\sum\log x_i$	n/θ^2

Moments: Exp(rate λ) $\mathbb{E}X^2 = 2/\lambda^2$ • Gamma(a,b) $\mathbb{E}X^2 = \frac{a(a+1)}{b^2}$, $\mathbb{E}\frac{1}{X} = \frac{b}{a-1}$ • Poisson $\mathbb{E}X^2 = \lambda+\lambda^2$ • N: $\mathbb{E}(X-\mu)^2 = \sigma^2$, $\mathbb{E}(X-\mu)^4 = 3\sigma^4$ • $\theta x^{\theta-1}$: $\mathbb{E}[-\log X] = \frac{1}{\theta}$ • Pareto $\alpha x_m^\alpha x^{-\alpha-1}$: $\mathbb{E}\log\frac{X}{x_m} = \frac{1}{\alpha}$ • lognormal $\mathbb{E}Y^k = e^{km+k^2\sigma^2/2}$.

Traps: course $\mathcal{I}$ already has n (don't divide again) • use the parameter named (rate/mean, precision/variance) • ℓ'' with data needs $\mathbb{E}$.

2 • Binomial regression (logit) and CIs

$Y_i \sim \mathrm{Bin}(m_i, p_i)$ indep., $g(p_i) = \eta_i = x_i^\top\beta$ (name dist., independence, link, η). $\ell(\beta) = c + \sum_i[y_i\eta_i - m_i\log(1+e^{\eta_i})]$; score $\sum_i(y_i - m_i p_i)x_i$; $\mathcal{I}(\beta) = J(\beta) = X^\top WX$, $W = \mathrm{diag}(m_i p_i(1-p_i))$ (canonical: no y in Hessian; probit/cloglog $J \ne \mathcal{I}$).

Links $\eta = g(p) \leftrightarrow p = g^{-1}(\eta)$: logit $\log\frac{p}{1-p} \leftrightarrow \frac{1}{1+e^{-\eta}}$ ($p = \frac12$ at $x = -\beta_0/\beta_1$, slope there $\beta_1/4$) • probit $\Phi^{-1}(p) \leftrightarrow \Phi(\eta)$ • cloglog $\log(-\log(1-p)) \leftrightarrow 1-\exp(-e^\eta)$ (asymmetric; "≥ 1 Poisson event").

CI for p (or μ) at x_0: (1) $\hat\eta_0 = x_0^\top\hat\beta$ (+offset); (2) se$= \sqrt{x_0^\top V x_0}$, V=**vcov**: $V_{00} + 2x_0V_{01} + x_0^2V_{11}$ (all variances + 2× every covariance); (3) $g^{-1}(\hat\eta_0 \pm 1.96\,\mathrm{se})$. Never $\hat p \pm 1.96\,\mathrm{se}(\hat p)$; asymmetric is correct.

Odds ratio for $+\Delta$ in x_j: $e^{\beta_j\Delta}$, CI $\exp\{\Delta(\hat\beta_j \pm 1.96\,\mathrm{se}_j)\}$; %change $= (OR-1)100$. Two non-baseline levels: $\mathrm{Var}(\hat\beta_a - \hat\beta_b) = V_{aa} + V_{bb} - 2V_{ab}$. e^β = OR only for logit (log link: rate ratio). Case–control: logit only shifts intercept ⇒ ORs valid, $\hat p$ not population risks.

Say: *CI on η then transform: $\hat\eta_0$ is linear in $\hat\beta$ so approx. normal; g^{-1} monotone keeps 95% coverage and p in $(0,1)$.* **Say:** *Offset: expected count $\propto$ exposure, $\mu_i = t_i\lambda_i$, so $\log t_i$ enters with coefficient 1; e^{β_j} = rate ratio per unit exposure.*

3 • Deviance, LRT, AIC (every GLM)

$D = 2[\ell_{\text{sat}} - \ell(\hat\beta)]$ on $n-p$ df; scaled D/ϕ (= D binomial/Poisson); $0\log 0 = 0$. Binomial $d_i = 2[y_i\log\frac{y_i}{\hat\mu_i} + (m_i-y_i)\log\frac{m_i-y_i}{m_i-\hat\mu_i}]$ • Poisson $d_i = 2[y_i\log\frac{y_i}{\hat\mu_i} - (y_i-\hat\mu_i)]$ (y_i=0: $2\hat\mu_i$) • Normal $(y_i-\hat\mu_i)^2$ • Gamma $-2[\log\frac{y_i}{\hat\mu_i} - \frac{y_i-\hat\mu_i}{\hat\mu_i}]$.

Counting: n = null df+1; $p = n-$resid df (incl. intercept; k-level factor $= k-1$; offset not counted).

GOF ($\phi = 1$ only): $D \approx \chi^2_{n-p}$ if all $m_i\hat p_i, m_i(1-\hat p_i) \ge 5$ (Poisson $\hat\mu_i \ge 5$). Invalid for 0/1 or sparse data, or ϕ estimated. $\mathbb{E}\chi^2_k = k$, so $D/\mathrm{df} \approx 1$ is reassuring.

LRT A⊂B: $D_A - D_B \approx \chi^2_s$, $s = \mathrm{df}_A - \mathrm{df}_B$; fine for 0/1 data. Factor = one block on $k-1$ df (dummy zs don't answer "does it matter"). Wald $z = \hat\beta/\mathrm{se}$, $z^2 \approx$LR; disagree ⇒ trust LRT. Different links/non-nested: AIC only.

AIC $= 2p - 2\hat\ell$, smaller better; $\mathrm{AIC}_B - \mathrm{AIC}_A = 2s + D_B - D_A$ (extra term worth it iff $\Delta D > 2$ per param).

$$\hat\ell = p - \tfrac{\mathrm{AIC}}{2}, \qquad \ell_{\text{sat}} = \hat\ell + \tfrac{D}{2}$$

(ℓ_{sat} is the same for every model on the data: a check.) Gamma/normal: R counts ϕ, $\hat\ell = (p+1) - \mathrm{AIC}/2$. Quasi: AIC NA (use Poisson fit).

Say: *Adequate: $D =$ _ on $n-p$ df $< \chi^2_{n-p}(.95) =$ _; valid as all fitted counts ≥ 5, so no evidence of lack of fit. (Sparse/0-1 data: test not valid, say so.)*

Say: *Prefer B: LRT $D_A - D_B =$ _ on s df $> \chi^2_s(.95) =$ _ ($p =$ _), reject H_0: extra coefs = 0; and $\mathrm{AIC}_B < \mathrm{AIC}_A$.* ("two quantitative reasons")

4 • R output decoder (read only)

`summary(glm(cbind(y,m-y)~x+f, family=binomial))`	R models proportions y/m with weight m. **Estimate. Std. Error** = $\hat\beta_j$, $\sqrt{V_{jj}}$ (link scale) • **z value**= $\hat\beta/\mathrm{se}$, **Pr(>|z|)** two-sided ($\phi = 1$ families) • **t value/Pr(>|t|)**: ϕ estimated (quasi, Gamma, normal), $n-p$ df • **fB** = dummy, level B vs baseline (first level) • **(Dispersion parameter … taken to be 1)**: ϕ *assumed*; quasi/Gamma print Pearson $\hat\phi = X^2/(n-p)$ • **Null deviance:** D_0 **on** $n-1$ (intercept only) • **Residual deviance:** D **on** $n-p$ • **AIC**= $2p - 2\hat\ell$ (NA quasi) • **Fisher Scoring iterations** = IWLS steps.
`anova(A,B,test="Chisq")`	rows: **Resid. Df** $n-p_A$, $n-p_B$; **Resid. Dev** D_A, D_B; **Df**= s, **Deviance**= $D_A - D_B$, **Pr(>Chi)**= $P(\chi^2_s > \Delta D)$
`anova(A,B,test="F")`	+ **F**= $(\Delta D/s)/\hat\phi_B$, **Pr(>F)** on $(s, n-p_B)$; on quasi fits (or Poisson fit + **dispersion=phihat**: ∞ denom. df)
`anova(m,test=..)`	sequential (order matters): row x = drop in D when x *added* to terms above; **NULL** row = null deviance
`drop1(m,test="Chisq")`	row x = model *without* x: its **Deviance**, **AIC**; **LRT**= $D_{-x} - D_{\text{full}}$; **<none>** = full model. Slides: drop1/step don't work for quasi (F uses $D/(n-p)$): quote **anova**
`predict(m,sd, type="link",se.fit=TRUE)`	**$fit**= $\hat\eta_0$, **$se.fit**=se($\hat\eta_0$) (quasi: already ×$\sqrt{\hat\phi}$; **$residual.scale**= $\sqrt{\hat\phi}$). CI: g^{-1}(fit ± 1.96 se.fit)
`type="response"`	$\hat\mu_0 = g^{-1}(\hat\eta_0)$; its se is delta-method: don't build the CI from it
`pchisq(D,df,lower.tail=F)`	upper-tail p (GOF if D = resid dev); **qchisq(.95,df)**, **qf(.95,s,d)** = 5% crit.
`sum(residuals(m, "pearson")^2)/ m$df.residual`	$\hat\phi = X^2/(n-p)$; two printed ⇒ F uses bigger model's
`fitted(m); vcov(m)`	$\hat\mu_i$ (binomial: $\hat p_i$, so $m_i\hat p_i$ = expected count); $V = \mathcal{I}(\hat\beta)^{-1}$
`offset(log(t))`	$\log\mu_i = \log t_i + x_i^\top\beta$, coef fixed at 1, not in p; e^{β_j} = rate ratios
`hatvalues; rstudent; cooks.distance`	h_{ii}; jack-knife residual; Cook's C_i

5 • What to look for where

Cue	Compute	Where
$\tilde\ell$ fitted/saturated	p − AIC/2; then $+D/2$	AIC, resid dev, df
n, p	null df+1; $n-$resid df	deviance lines
model adequate?	D vs $\chi^2_{n-p}(.95)$; fitted counts ≥ 5?	resid dev, **qchisq**
better model, 2 reasons	LRT ΔD vs χ^2_s; AIC	2 summaries/anova
evidence of OD	$D/\mathrm{df} \gg 1$; $\hat\phi \gg 1$; GOF p tiny	resid dev, phi line
SE allowing for OD	se×$\sqrt{\hat\phi}$ (that model's)	Std. Error, phi
Poisson SE from quasi	quasi se ÷$\sqrt{\hat\phi}$; estimate same	quasi summary
quasi t value	$z/\sqrt{\hat\phi} = \hat\beta/(\mathrm{se}\sqrt{\hat\phi})$	
F (hidden row)	$(\Delta D/s)/\hat\phi_B$ vs $F_{s,n-p_B}$	anova + big $\hat\phi$
predicted mean/prob, CI	$\hat\eta$ by hand (offset!), CI on η, then g^{-1}	coefs+vcov / se.fit
effect of $+\Delta x$	$e^{\beta\Delta}$; CI via endpoints	Estimate, se
rate per t units	$\hat\mu = t\,e^{x^\top\beta}$ (set t in offset)	newdata

6 • Exponential family, canonical link, $v(\mu)$ (8–10 marks)

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right], \qquad a(\phi) = \phi$$

$$\mathbb{E}Y = b'(\theta), \qquad \mathrm{Var}\,Y = a(\phi)b''(\theta) = a(\phi)v(\mu)$$

1. Take logs; **say support doesn't depend on the parameter** (Unif$(0,\theta)$, shifted exp: not EF).
2. Term $y \times h$(param): $\theta = h$ (param must multiply y, not y^2, $\log y$, e^y).
3. Factor scale out of y-term & param-only term: $a(\phi)$.
4. Param-only term $= -b/a(\phi)$: **rewrite b in θ** (invert $\theta(\mu)$). Rest $= c(y,\phi)$ (known consts OK).
5. Check $b'(\theta) = \mu$. $v(\mu) = b''(\theta)$ at $\theta = \theta(\mu)$; $\mathrm{Var}\,Y = a(\phi)v(\mu)$ ($\ne v$ unless $\phi = 1$).
6. Canonical link $g = (b')^{-1}$: solve $\mu = b'(\theta)$, $g(\mu) = \theta(\mu)$.

dist	$\theta(\mu)$ = canon. g	$b(\theta)$	ϕ	$v(\mu)$
$N(\mu,\sigma^2)$	μ	$\theta^2/2$	σ^2	1
Poisson	$\log\mu$	e^θ	1	μ
Bernoulli	$\log\frac{\mu}{1-\mu}$	$\log(1+e^\theta)$	1	$\mu(1-\mu)$
Bin(m) count	$\log\frac{\mu}{m-\mu}$	$m\log(1+e^\theta)$	1	$\mu(1-\frac{\mu}{m})$
Bin prop. Y/m	logit μ	$\log(1+e^\theta)$	$1/m$	$\mu(1-\mu)$
Exp(rate λ)	$-1/\mu$	$-\log(-\theta)$	1	μ^2
Gamma(μ,ν)	$-1/\mu$	$-\log(-\theta)$	$1/\nu$	μ^2
Inv. Gauss.	$-1/(2\mu^2)$	$-\sqrt{-2\theta}$	$1/\lambda$	μ^3
Geom. $p(1-p)^y$	$\log\frac{\mu}{1+\mu}$	$-\log(1-e^\theta)$	1	$\mu(1+\mu)$
NegBin, r known	$\log\frac{\mu}{\mu+r}$	$-r\log(1-e^\theta)$	1	$\mu+\frac{\mu^2}{r}$
mean of k Pois.	$\log\mu$	e^θ	$1/k$	μ

Natural θ: Exp $-\lambda$; Geom $\log(1-p)$; Bernoulli logit p. Gamma(μ,ν) = mean, shape; R's "inverse" link is $1/\mu$. Gamma: $\log f = \frac{y(-1/\mu)-\log\mu}{1/\nu} + \nu\log\nu + (\nu-1)\log y - \log\Gamma(\nu)$. Lemma proof: $\mathbb{E}\ell' = 0$, $\mathbb{E}\ell'' = -\mathbb{E}(\ell')^2$ (one obs).

Canonical link ⇒ $\ell = \sum[y_i x_i^\top\beta - b(x_i^\top\beta)]/\phi + c$: $X^\top y$ sufficient; score $X^\top(y-\mu)/a(\phi)$; Hessian has no y so $J = \mathcal{I}$, Newton–Raphson = Fisher scoring. Log link often preferred over $-1/\mu$, $\log\frac{\mu}{\mu+r}$: any $\eta \in \mathbb{R}$ valid, multiplicative effects.

7 • GLM fitting (IWLS), unknown ϕ, diagnostics

GLM: indep. Y_i from one EF (common ϕ), $\eta_i = x_i^\top\beta$, monotone differentiable $g(\mu_i) = \eta_i$; ϕ not modelled.

$$U_j = \sum_i \frac{(y_i-\mu_i)x_{ij}}{a(\phi)v(\mu_i)g'(\mu_i)}, \qquad W_{ii} = \frac{1}{g'(\mu_i)^2 v(\mu_i)}$$

$$z_i = \eta_i + (y_i-\mu_i)g'(\mu_i), \qquad \hat\beta^{\text{new}} = (X^\top WX)^{-1}X^\top Wz$$

Start $\hat\mu = y$; iterate (= Fisher scoring). $\mathrm{Var}\,\hat\beta = a(\phi)(X^\top\hat WX)^{-1}$: ϕ cancels in update, not variance ⇒ SE$\propto\sqrt{\hat\phi}$. Canonical $W_{ii} = v(\mu_i)$ (Poisson-log μ_i, logit $m_i p_i(1-p_i)$); Gamma-log $W = 1$. $W^{-1} \propto \mathrm{Var}\,z_i$ (delta method).

ϕ **unknown** (Gamma $\mathrm{Var}\,Y = \phi\mu^2$, CV= $\sqrt{\phi}$; normal): $\hat\phi = X^2/(n-p)$, $X^2 = \sum(y_i-\hat\mu_i)^2/v(\hat\mu_i)$ (Pearson, *not* $D/(n-p)$); t tests, F tests, no deviance GOF ($\hat\phi$ uses same residuals).

Residuals: Pearson $r_P = \frac{y-\hat\mu}{\sqrt{v(\hat\mu)}}$, $\sum r_P^2 = X^2$ • deviance $r_D = \mathrm{sign}(y-\hat\mu)\sqrt{d_i}$, $\sum r_D^2 = D$ (R default) • standardised $r/\sqrt{(1-h_{ii})\hat\phi}$ • jack-knife **rstudent**: large ⇒ outlier. **Leverage** h_{ii}: diag of IWLS hat matrix, $\sum h_{ii} = p$, flag $> 2p/n$ (potential only).

Cook's $C_i \approx \frac{r_{Pi}^2 h_{ii}}{p\hat\phi(1-h_{ii})^2}$: needs leverage *and* residual; refit without i to confirm. **Plots** r_D vs $\hat\eta$: trend ⇒ wrong mean/link; changing spread ⇒ wrong $v(\mu)$; bands = discrete y. Half-normal: tail jump.

8 • Overdispersion and quasi-likelihood

Def: actual $\mathrm{Var}\,Y_i >$ model's (Poisson μ_i; binomial $m_i p_i(1-p_i)$). Causes: dependent trials/events (clustering); p or rate varying between units (missing covariates). Large D can also be wrong mean structure/outliers: check residuals vs $\hat\eta$ first. 0/1 data can't show it.

Quantitative reasons: (i) resid $D \gg$ resid df; (ii) GOF **pchisq(D,df,lower=F)** tiny (counts ≥ 5); (iii) $\hat\phi = X^2/(n-p) \gg 1$.

Say: *Ignoring OD: $\hat\beta$ unaffected (ϕ cancels from score) but true $\mathrm{Var}\,\hat\beta$ is $\phi\times$ reported: SEs too small, z too big, p-values too small, CIs too narrow, χ^2 LRTs too significant ⇒ predictors wrongly declared significant/kept.*

Quasi-likelihood: specify only $g(\mu_i) = x_i^\top\beta$, $\mathbb{E}Y_i = \mu_i$, $\mathrm{Var}\,Y_i = \phi v(\mu_i)$ (IWLS needs only g, v). $u_i = \frac{Y_i-\mu_i}{\phi v(\mu_i)}$: $\mathbb{E}u_i = 0$, $\mathrm{Var}\,u_i = \frac{1}{\phi v}$ $= -\mathbb{E}\frac{\partial u_i}{\partial\mu_i}$ (acts like a score).

$$Q_i = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi v(t)}\,dt \le 0, \qquad Q = \sum_i Q_i$$

$$\frac{D_Q}{\phi} = -2\sum_i Q_i \;(Q_{\text{sat}} = 0), \qquad D_Q = -2\sum_i \int_{y_i}^{\mu_i} \frac{y_i - t}{v(t)}\,dt$$

$v(\mu)$	ϕQ_i (definite integral; slides drop μ-free terms)	$D_Q =$
μ (q-Poisson)	$y\log\frac{\mu}{y} - (\mu - y)$ [slides: $\frac{1}{\phi}(y\log\mu - \mu)$]	Poisson dev.
$\frac{\mu(m-\mu)}{m}$ (q-binomial)	$y\log\frac{\mu}{y} + (m-y)\log\frac{m-\mu}{m-y}$	binomial dev.
μ^2	$-\frac{y}{\mu} + 1 - \log\frac{\mu}{y}$	Gamma dev.

$\hat\phi = X^2/(n-p)$ (moment est.; Q has no finite max in ϕ). Consistent, asympt. normal, less efficient than the true MLE.

Quasi-Poisson/binomial output: same $\hat\beta$, fitted values, predictions, deviances; $\mathrm{SE}_q = \sqrt{\hat\phi}\,\mathrm{SE}_{\phi=1}$; $t = z/\sqrt{\hat\phi}$; AIC NA; no GOF; compare nested models by

$$F = \frac{(D_A - D_B)/s}{\hat\phi_B} \approx F_{s,n-p_B}, \qquad \hat\phi_B = \frac{X_B^2}{n-p_B}$$

($\hat\phi$ from the bigger model B: $\hat\phi_A$ is inflated by the lack of fit being tested; F not χ^2 because ϕ is estimated: $\frac{\chi^2_s/s}{\chi^2_{n-p}/(n-p)}$.)

Say: *Poisson: $\mathrm{Var}\,Y_i = \mu_i$ ($\phi = 1$), full distribution. Quasi-Poisson: only $\mathbb{E}Y_i = \mu_i$, $\mathrm{Var}\,Y_i = \phi\mu_i$, ϕ estimated by $X^2/(n-p)$. Same $\hat\beta$ and deviance; SEs ×$\sqrt{\hat\phi}$; F tests not χ^2; no likelihood so no AIC.*

Say: *Which model? χ^2 LRT keeps the term ($p =$ _) but with $\hat\phi =$ _, $F =$ _ on $(s, n-p_B)$, $p >$ 0.05: allowing for OD prefer the smaller model.*

Say: *Non-standard response by quasi? Yes: link g keeping μ in range, $\mathrm{Var}\,Y = \phi v(\mu)$ from the mean–variance relation, fit by IWLS* (**quasi(link=,variance=)**), $\hat\phi = X^2/(n-p)$, *SEs*×$\sqrt{\hat\phi}$, *F tests.*

Alternatives: negative binomial ($\mathrm{Var} = \mu + \mu^2/r$); Bayesian: unit random effect λ_i ~Gamma (marginal NegBin, $\mathrm{Var}\,y_i = \mathbb{E}\lambda_i + \mathrm{Var}\,\lambda_i > \mathbb{E}y_i$) or $\log\lambda_i = x_i^\top\beta + \varepsilon_i$.

Traps: two $\hat\phi$ printed: F uses big model's, SE uses that model's own • F df $(s, n-p_B)$ • never rescale the estimate • $\hat\phi$ is Pearson, not D/df.

16 • Reading R code: argument traps

`rnorm(n,mean,sd)`	**sd** not var: $N(\frac{B}{A}, \frac{1}{A})$ = **rnorm(1,B/A,sqrt(1/A))**
`rgamma(n,shape,rate)`	3rd positional = **rate** (scale = 1/rate); **1/rgamma(1,a,b)** = IG(a,b)
`dnorm(x,m,sd,log=TRUE)`	log f; sums = log-lik; differences = log ratios
`exp(lpost(prop)-lpost(cur))`	π^*/π^c; no q term ⇒ symmetric proposal
`if(runif(1)<ratio)`	accept w.p. $\min\{1, \text{ratio}\}$ (or **log(runif(1))<lr**)
`runif(1)<=f(x)/M`	rejection accept (uniform g)
`sample(1:n,1,prob=w)`	discrete draw; w normalised internally
`cumsum(y); mean(th>c)`	c_m; MC estimate of $P(\theta > c\|x)$
`1-mean(duplicated(ch))`	acceptance rate (also **mean(diff(x)!=0)**)
`apply(chs,2,sd); sd(c(chs))`	within-chain sds (W = mean); pooled B
`acf(x,plot=F)$acf[2]; ch[-(1:b),]`	$\hat\rho_1$; drop burn-in
`proposal=rnorm(3,mean=param, sd=c(.1,.5,.3))`	symmetric RW, one sd per parameter ⇒ q-ratio 1
`posterior(p)=likelihood(p)+prior(p)`	log target = ∑**dnorm(y,pred,sd,log=T)** + log priors
`if(runif(1)<probab) chain[i+1,]= proposal else chain[i,]`	accept, else repeat current value
`rgamma(1,a1+cfreq[M],b1+M)`	coal Gibbs: $\lambda_1\|\cdot$ ~Ga(shape, rate)
`sample(1:n,1,prob=(la1/la2)^ cfreq*exp(la2-la1)*(1:n))`	discrete conditional of M over $m = 1..n$
`x<-runif(1,a,b); y<-runif(1,0,K); if(y<fx(x)) return(x)`	rejection with uniform envelope height K

17 • Distribution facts NOT on the formula sheet

Bernoulli(p) $p^y(1-p)^{1-y}$, $\mathbb{E}=p$, Var $=p(1-p)$ • Bin(m,p) $\binom{m}{y}p^y(1-p)^{m-y}$, mp, $mp(1-p)$ • Geometric (fails) $p(1-p)^y$: $\mathbb{E}=\frac{1-p}{p}$, Var $=\frac{1-p}{p^2}$ (trials $\theta(1-\theta)^{x-1}$: $\mathbb{E}=\frac{1}{\theta}$) • NegBin$(\mu,r)$ $\frac{\Gamma(y+r)}{\Gamma(r)y!}(\frac{r}{r+\mu})^r(\frac{\mu}{r+\mu})^y$, Var $=\mu+\frac{\mu^2}{r}$; Poisson–Gamma(a,b) marginal $\frac{\Gamma(a+y)}{\Gamma(a)y!}\frac{b^a}{(b+1)^{a+y}}$, $\mathbb{E}=\frac{a}{b}$, Var $=\frac{a}{b}+\frac{a}{b^2}$ • Beta(α,β) $\frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha,\beta)}$, $B=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$, $\mathbb{E}=\frac{\alpha}{\alpha+\beta}$, Var $=\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ • Multinomial $\frac{n!}{\prod n_k!}\prod\pi_k^{n_k}$, $\mathbb{E}n_k=n\pi_k$, Var $=n\pi_k(1-\pi_k)$ • Dirichlet$(\alpha)\propto\prod\pi_k^{\alpha_k-1}$, $\mathbb{E}\pi_k=\alpha_k/\sum\alpha$ • **Variances:** Normal ω^2, Exp $1/\lambda^2$, Gamma a/b^2, Inv-Gamma $\frac{b^2}{(a-1)^2(a-2)}$ $(a>2)$, Poisson λ, Categorical(π_1,π_2) $\pi_1\pi_2$ • χ^2_k $=$Ga$(\frac{k}{2},\frac{1}{2})$: mean k, var $2k$ • $\Gamma(n)=(n-1)!$, $\Gamma(\frac{1}{2})=\sqrt{\pi}$ • $\mathbb{E}X=\mathbb{E}[\mathbb{E}(X|Z)]$, Var $X=\mathbb{E}[\text{Var}(X|Z)]+\text{Var}[\mathbb{E}(X|Z)]$.

18 • 5% critical values; general traps

$\chi^2_{df}(.95)$: 1: 3.84, 2: 5.99, 3: 7.81, 4: 9.49, 5: 11.07, 6: 12.59, 7: 14.07, 8: 15.51, 9: 16.92, 10: 18.31, 20: 31.41, 25: 37.65, 30: 43.77, 50: 67.50. $F_{s,d}(.95)$ for d =10, 20, 30, 60, ∞: s=1: 4.96, 4.35, 4.17, 4.00, 3.84; s=2: 4.10, 3.49, 3.32, 3.15, 3.00; s=3: 3.71, 3.10, 2.92, 2.76, 2.60. z: 1.645 (90%), 1.96 (95%), 2.576 (99%).

Traps: sheet Normal is (mean, *variance*): convert precision/sd • Gamma rate vs scale • IG mean $\frac{b}{a-1}$; $1/\mathbb{E}\tau\neq\mathbb{E}\sigma^2$ • always give name *and* parameters • explain parts: 1–2 sentences quoting the number that justifies it.

9 • Mixtures and EM (K–12 marks): Q, E-step, M-step

$Z_i\sim$Cat$(\pi_1..\pi_K)$ (K=2: $\pi,1-\pi$), $X_i|Z_i{=}k\sim f(\cdot;\theta_k)$; pick component then sample (not $\pi_1X_1+\pi_2X_2$). **Incomplete** $\ell(\theta)=\sum_i\log\sum_k\pi_kf(x_i;\theta_k)$: log of a sum, no closed form.

$$P(X,Z\mid\theta)=\prod_i\prod_k[\pi_kf(x_i;\theta_k)]^{I(Z_i=k)}$$

$$\log P(X,Z\mid\theta)=\sum_i\sum_k I(Z_i{=}k)[\log\pi_k+\log f(x_i;\theta_k)]$$

$$Q(\theta,\theta^0)=E_{Z|X,\theta^0}[\log P(X,Z|\theta)]$$
$$=\sum_i\sum_k E[I(Z_i{=}k)|X,\theta^0][\cdots]$$
$$=\sum_i\sum_k\gamma_{ik}[\log\pi_k+\log f(x_i;\theta_k)]$$

(line 2 = complete log-lik; line 4 by linearity; line 5 since $E[I(Z_i{=}k)|X,\theta^0]=P(Z_i{=}k|X_i,\theta^0)$.) Expand $\log f$: Normal $-\frac{1}{2}\log(2\pi\sigma_k^2)-\frac{(x_i-\mu_k)^2}{2\sigma_k^2}$ • Poisson $x_i\log\lambda_k-\lambda_k-\log x_i!$ • Exp $\log\lambda_k-\lambda_kx_i$ • Bin(m) $x_i\log p_k+(m-x_i)\log(1-p_k)$+c. K=2: $\gamma_{i1}\log\pi+\gamma_{i2}\log(1-\pi)$.

E-step (Bayes' rule at θ^0): $\gamma_{ik}=P(Z_i{=}k|X_i,\theta^0)=\frac{\pi_k^0f(x_i;\theta_k^0)}{\sum_j\pi_j^0f(x_i;\theta_j^0)}$, $\sum_k\gamma_{ik}=1$, $N_k=\sum_i\gamma_{ik}$, $\sum_kN_k=n$. Cancel only factors common to all terms ($x!$, $\frac{1}{\sqrt{2\pi}}$; $\frac{1}{\sigma}$ only if equal). K=2: $\gamma_{i1}=1/(1+e^d)$, $d=\log\frac{\pi_2f_2}{\pi_1f_1}$.

Algorithm: init θ^0, $\ell(\theta^0)$ → E-step → M-step $\hat\theta=\arg\max_\theta Q(\theta,\theta^0)$ (γ fixed) → $\ell(\hat\theta)$; stop if $|\Delta\ell|<\varepsilon$ (incomplete ℓ, not Q) or max.iter; else $\theta^0\leftarrow\hat\theta$.

M-step (differentiate Q, set 0; γ = weights):

π_k (any)	$\hat\pi_k=N_k/n$. K=2: $\frac{N_1}{\pi}-\frac{N_2}{1-\pi}=0$. $K\geq3$: Lagrange $\frac{N_k}{\pi_k}=\alpha$, $\alpha=n$
Normal $\mu_k;\sigma_k^2$	$\hat\mu_k=\frac{\sum_i\gamma_{ik}x_i}{N_k}$; $\hat\sigma_k^2=\frac{\sum_i\gamma_{ik}(x_i-\hat\mu_k)^2}{N_k}$ (diff. wrt $v=\sigma^2$, new $\hat\mu_k$)
shared σ^2	$\hat\sigma^2=\frac{1}{n}\sum_i\sum_k\gamma_{ik}(x_i-\hat\mu_k)^2$
means known (e.g. 0)	$\hat\sigma_k^2=\sum_i\gamma_{ik}(x_i-\mu_k)^2/N_k$; no μ update
Poisson λ_k	$\hat\lambda_k=\sum_i\gamma_{ik}x_i/N_k$
Poisson$(t_i\lambda_k)$	$\hat\lambda_k=\sum_i\gamma_{ik}x_i/\sum_i\gamma_{ik}t_i$ (E-step keeps $e^{-t_i\lambda_k}$)
Exp rate λ_k	$\hat\lambda_k=N_k/\sum_i\gamma_{ik}x_i$ (inverted!)
Bin(m,p_k)	$\hat p_k=\sum_i\gamma_{ik}x_i/(mN_k)$
tied Pois $\lambda/c\lambda$	$\hat\lambda=\sum_ix_i/\sum_i(\gamma_{i1}+c\gamma_{i2})$
tied Exp $\lambda/c\lambda$	$\hat\lambda=n/\sum_i(\gamma_{i1}+c\gamma_{i2})x_i$
$N(\mu,\sigma^2)$ / $N(\mu,c\sigma^2)$	$w_i=\gamma_{i1}+\gamma_{i2}/c$: $\hat\mu=\sum w_ix_i/\sum w_i$, $\hat\sigma^2=\frac{1}{n}\sum w_i(x_i-\hat\mu)^2$
ZIP: $Z{=}1\Rightarrow X{=}0$	$\gamma_{i1}=\frac{\pi^0}{\pi^0+(1-\pi^0)e^{-\lambda^0}}$ if $x_i{=}0$ else 0; $\hat\lambda=\frac{\sum x_i}{n-N_1}$

Twists (rewrite complete likelihood first, then Q, then M-step):
- $x_{m+1..n}$ *known from comp. 2, no* Z: Q += $\sum_j\log f(x_j;\theta_2)$, weight 1, no $\log\pi$. $\hat\mu_2=\frac{\sum_{i\leq m}\gamma_{i2}x_i+\sum_jx_j}{N_2+(n-m)}$; $\hat\pi=\frac{1}{m}\sum_{i\leq m}\gamma_{i1}$.
- *labels observed* $Z_j=2$: + $\sum_j[\log(1-\pi)+\log f(x_j;\theta_2)]$: same $\hat\mu_2$, $\hat\pi=\frac{1}{n}\sum_{i\leq m}\gamma_{i1}$.
- known/fixed parameter: a constant (no update) but still in γ • tied parameter: one derivative summing every component containing it.

Classify: $\arg\max_k\gamma_{ik}$; K=2: comp. 1 iff $\pi_1f_1(x)>\pi_2f_2(x)$; equal σ: boundary $\frac{\mu_1+\mu_2}{2}+\frac{\sigma^2\log(\pi_2/\pi_1)}{\mu_1-\mu_2}$; report γ as the confidence.

Moments: $\mathbb{E}X=\sum\pi_k\mu_k=\bar\mu$; Var $X=\sum\pi_k\sigma_k^2+\sum\pi_k(\mu_k-\bar\mu)^2$ (K=2: $+\pi_1\pi_2(\mu_1-\mu_2)^2$); Poisson mixture Var $X>\mathbb{E}X$.

Why EM never decreases ℓ ($q(Z)=P(Z|X,\theta^0)$, Jensen):

$$\ell(\theta)=\log\textstyle\sum_Z q(Z)\frac{P(X,Z|\theta)}{q(Z)}$$
$$\geq\textstyle\sum_Z q(Z)\log P(X,Z|\theta)-\sum_Z q(Z)\log q(Z)$$
$$=Q(\theta,\theta^0)+C\qquad(=\text{ at }\theta=\theta^0:\text{ ratio}=P(X|\theta^0))$$
$$\ell(\hat\theta)\geq Q(\hat\theta,\theta^0)+C\;\geq\;Q(\theta^0,\theta^0)+C=\ell(\theta^0)$$

Say: *E-step makes $Q(\cdot,\theta^0)+C$ a lower bound on ℓ touching it at θ^0; M-step raises Q, so ℓ cannot fall.* **Say:** *EM climbs to a stationary point near its start (local max): run several starts, keep the largest final incomplete ℓ. Same ℓ with swapped labels = label switching. Equal starts ($\mu_1^0{=}\mu_2^0$) give $\gamma_{ik}=\pi_k^0$, never separate. Hitting max.iter ≠ converged. Decreasing ℓ = bug.* **Say:** *Why Q: the log acts on each π_kf_k with fixed weights, so Q splits into weighted complete-data log-likelihoods with closed-form maximisers.*

Lecturer's EM code ($n\times K$ matrices: row i, column k)

`Z=sample(1:K,n,TRUE,prob=pi)`	$Z_i\sim$Cat(π)
`X=rnorm(n,mu[Z],sqrt(sigma2[Z]))`	$X_i\|Z_i$ (sd!)
`prob.x.z[,k]=dnorm(X,mu[k],sqrt(2))*pi[k]`	$\pi_kf_k(x_i)=P(x_i,Z_i{=}k)$
`rowSums(prob.x.z)`	$\sum_k\pi_kf_k(x_i)=P(x_i\|\theta)$
`P_ik=prob.x.z/rowSums(prob.x.z)`	γ_{ik} (E-step); colSums $=N_k$
`pi.new=colSums(P_ik)/sum(P_ik)`	$\hat\pi_k=N_k/n$
`mu.new=colSums(P_ik*X)/colSums(P_ik)`	$\hat\mu_k$
`sum(log(rowSums(prob.x.z)))`	incomplete ℓ
`while((delta.ll>epsilon)& (n.iter<max.iter))`	stop rule: `length(log_liks)`=iters+1
`apply(P_ik,1,which.max)`	hard classification

Traps: $\hat\pi=N_k/n$ not N_k/n_k • $\hat\sigma_k^2$ divides by N_k • γ uses θ^0, density in Q uses θ • never differentiate γ • forgetting π_k (or unequal $1/\sigma_k$) in γ • K=3: $1-\gamma_1\neq\gamma_2$.

10 • Bayes: kernels and conjugacy

$p(\theta|x)=\frac{p(x|\theta)p(\theta)}{p(x)}\propto p(x|\theta)p(\theta)$. Drop factors free of θ ($p(x)$, $x_i!$, $\sqrt{2\pi}$, $\frac{b^a}{\Gamma(a)}$); **keep** $\tau^{n/2}$, $\sqrt{\kappa_0\tau}$ when the target is τ. Joint prior $p(\mu,\tau)=p(\mu|\tau)p(\tau)$.

Kernels: $\tau^se^{-r\tau}$ →Gamma$(s{+}1,r)$, var $\frac{a}{b^2}$ • $(\sigma^2)^{-s}e^{-r/\sigma^2}$ →IG$(s{-}1,r)$, mean $\frac{b}{a-1}$ • $e^{-\frac{A}{2}\mu^2+B\mu}$ → $N(\frac{B}{A},\frac{1}{A})$ • $\theta^s(1-\theta)^r$ →Beta$(s{+}1,r{+}1)$ • $\lambda^{\sum y}e^{-n\lambda}$ →Gamma$(\sum y{+}1,n)$. $-\frac{A}{2}\mu^2+B\mu=-\frac{A}{2}(\mu-\frac{B}{A})^2+\frac{B^2}{2A}$. $\tau\sim$Ga$(a,b)\iff1/\tau\sim$IG(a,b). $\int_0^\infty\tau^{a-1}e^{-b\tau}=\frac{\Gamma(a)}{b^a}$; $\int e^{-\frac{A}{2}\mu^2+B\mu}=\sqrt{\frac{2\pi}{A}}e^{\frac{B^2}{2A}}$; $\int_0^1\theta^{\alpha-1}(1-\theta)^{\beta-1}=B(\alpha,\beta)$.

likelihood; prior	posterior ($S_\mu=\sum(x_i-\mu)^2$, $S=\sum(x_i-\bar x)^2$)
$N(\mu,\frac{1}{\tau})$, τ known; $\mu\sim N(m_0,\frac{1}{\kappa_0})$	$N(\frac{\kappa_0m_0+n\tau\bar x}{\kappa_0+n\tau},\frac{1}{\kappa_0+n\tau})$, precisions add
μ known; $\tau\sim$Ga(a,b) / $\sigma^2\sim$IG(a,b)	Ga / IG$(a+\frac{n}{2},\ b+\frac{S_\mu}{2})$
Poisson [exposure t_i]; Ga(a,b)	Ga$(a+\sum y_i,\ b+n)$ [$b+\sum t_i$]
Exp(rate λ); Ga(a,b)	Ga$(a+n,\ b+\sum x_i)$
Bin(m_i,θ); Beta(α,β)	Beta$(\alpha+\sum y_i,\ \beta+\sum(m_i-y_i))$
Geometric (trials) $\theta(1-\theta)^{x-1}$; Beta	Beta$(\alpha+n,\ \beta+\sum x_i-n)$
Multinomial counts n_k; Dirichlet(α)	Dirichlet(α_k+n_k)
$\mu\|\tau\sim N(m_0,\frac{1}{\kappa_0\tau})$, $\tau\sim$Ga(a,b)	$\mu\|\tau,x\sim N(\frac{\kappa_0m_0+n\bar x}{\kappa_0+n},\frac{1}{(\kappa_0+n)\tau})$; $\tau\|x\sim$Ga$(a+\frac{n}{2},\ b+\frac{S}{2}+\frac{n\kappa_0(\bar x-m_0)^2}{2(n+\kappa_0)})$

Shrinkage: post. mean $=w$ (prior mean)+$(1-w)$ MLE; Normal $w=\frac{\kappa_0}{\kappa_0+n\tau}$, Poisson–Gamma $w=\frac{b}{b+n}$; $n\to\infty$: $\to N(\hat\theta,\cdot)$, prior irrelevant; post. var < prior var and < MLE var. **Summaries:** mean; 95% credible = central (Normal: mean±1.96sd). Predictive $p(\tilde x|x)=\int p(\tilde x|\theta)p(\theta|x)d\theta$; Normal $N(m_n,\frac{1}{\kappa_n}+\frac{1}{\tau})$; simulate $\theta^{(t)}$, then $\tilde x^{(t)}|\theta^{(t)}$.

Say: *Credible: $P(\theta\in I|x)=0.95$, θ random, data fixed. Confidence: over repeated samples 95% of intervals cover the fixed θ.* **Say:** *Vague prior: proper, nearly flat ($N(0,10^6)$, $Ga(.001,.001)$), data dominate. Informative: real prior knowledge, κ_0 comparable to $n\tau$, shifts/narrows the posterior. Improper ($p(\mu)\propto1$, $p(\tau)\propto\frac{1}{\tau}$): $\int=\infty$, must check the posterior is proper.* $p(\tau)\propto\frac{1}{\tau}\Rightarrow$Ga$(\frac{n}{2},\frac{S}{2})$ ($S>0$); Poisson $p(\lambda)\propto\frac{1}{\lambda}\Rightarrow$Ga$(\sum y,n)$ needs $\sum y>0$; $p(\mu)\propto1\Rightarrow N(\bar x,\frac{1}{n\tau})$. BUGS `dnorm(m,prec)`: precision.

11 • Gibbs sampler: full conditionals

(1) Joint = likelihood × every prior (each once). (2) For θ_j: others constant; keep every factor containing θ_j. (3) Kernel: quadratic in exp → Normal; $\theta^se^{-r\theta}$ → Gamma; finite set → weights ÷ total. (4) State **name + both parameters**. Scheme: initial values; for $t=1..T$ draw each $\theta_j\sim p(\theta_j|\theta_{-j},x)$ using the *newest* values; discard burn-in; average. = MH with $\alpha=1$.

model ($x_i\sim N(\mu,\frac{1}{\tau})$)	conditionals
indep. $\mu\sim N(m_0,\frac{1}{\kappa_0})$, $\tau\sim$Ga(a,b)	$\mu\|\cdot\sim N(\frac{\kappa_0m_0+\tau\sum x_i}{\kappa_0+n\tau},\frac{1}{\kappa_0+n\tau})$; $\tau\|\cdot\sim$Ga$(a+\frac{n}{2},b+\frac{1}{2}\sum(x_i-\mu)^2)$
joint $\mu|\tau\sim N(m_0,\frac{1}{\kappa_0\tau})$	$\mu\|\cdot\sim N(\frac{\kappa_0m_0+n\bar x}{\kappa_0+n},\frac{1}{(\kappa_0+n)\tau})$; $\tau\|\cdot\sim$Ga$(a+\frac{n+1}{2},b+\frac{1}{2}[\sum(x_i-\mu)^2+\kappa_0(\mu-m_0)^2])$
+ source $y_{1..m}\sim N(k\mu,\frac{1}{c\tau})$	μ: $A=\kappa_0+(n+ck^2m)\tau$, $B=\kappa_0m_0+\tau(\sum x_i+ck\sum y_j)$, $N(\frac{B}{A},\frac{1}{A})$; τ: Ga$(a+\frac{n+m}{2},b+\frac{1}{2}[\sum(x_i-\mu)^2+c\sum(y_j-k\mu)^2])$
$\sigma^2\sim$IG(a,b)	IG$(a+\frac{n}{2},b+\frac{1}{2}\sum(x_i-\mu)^2)$; R `1/rgamma(1,..)`
hierarchical $\theta_j|\mu\sim N(\mu,\frac{1}{\tau_\theta})$, $y_{ij}\sim N(\theta_j,\frac{1}{\tau})$	$\mu\|\cdot\sim N(\frac{\kappa_0m_0+\tau_\theta\sum_j\theta_j}{\kappa_0+J\tau_\theta},\frac{1}{\kappa_0+J\tau_\theta})$ (no y); $\theta_j\|\cdot\sim N(\frac{n_j\tau\bar y_j+\tau_\theta\mu}{n_j\tau+\tau_\theta},\frac{1}{n_j\tau+\tau_\theta})$; $\tau_\theta\sim$Ga$(c+\frac{J}{2},d+\frac{1}{2}\sum_j(\theta_j-\mu)^2)$
Poisson changepoint M ($c_m=\sum_{i\leq m}y_i$)	$\lambda_1\|\cdot\sim$Ga(a_1+c_M,b_1+M), $\lambda_2\|\cdot\sim$Ga$(a_2+c_n-c_M,b_2+n-M)$, $P(M{=}m\|\cdot)\propto(\frac{\lambda_1}{\lambda_2})^{c_m}e^{(\lambda_2-\lambda_1)m}$
Bin(N,p), $p\sim$Beta, $N\sim$Poi(λ)	$p\|\cdot\sim$Beta$(\alpha+x,\beta+N-x)$; $N-x\|\cdot\sim$Poi$(\lambda(1-p))$

Say: *Gibbs limitation: strong posterior correlation ⇒ narrow conditionals, one-coordinate moves zig-zag slowly, high autocorrelation, small ESS (MH can propose joint moves); needs conditionals you can sample.*

12 • Metropolis–Hastings (the α ledger)

Propose $\theta^*\sim q(\cdot|\theta^c)$; $u\sim U(0,1)$; accept if $u<\alpha$, else repeat θ^c. $\pi=p(x|\theta)p(\theta)$ unnormalised ($p(x)$ cancels).

$$\alpha=\min\left\{1,\ \frac{\pi(\theta^*)}{\pi(\theta^c)}\cdot\frac{q(\theta^c|\theta^*)}{q(\theta^*|\theta^c)}\right\}\quad\text{(reverse/forward)}$$

Ledger (one line per factor, 4 dp, round once): (i) likelihood ratio incl. $(\tau^*/\tau^c)^{n/2}$; (ii) each prior ratio (keep $\sqrt{\tau}$ of $p(\mu|\tau)$); (iii) proposal ratio per proposed parameter; multiply; min$\{1,\cdot\}$. Logs: accept iff $\log u<\log\pi^*-\log\pi^c+\log q(c|*)-\log q(*|c)$. **Reverse density** starts at θ^*, lands on θ^c: a rate/variance set by θ^c going forward is set by θ^* in reverse. E.g. Exp(rate $1/\tau^c$): ratio $=\frac{(1/\tau^*)e^{-\tau^c/\tau^*}}{(1/\tau^c)e^{-\tau^*/\tau^c}}$. Symmetric RW ($N(\theta^c,s^2)$, $U[\theta^c\pm\epsilon]$): ratio 1, $\alpha=\min\{1,\frac{\pi^*}{\pi^c}\}$ • independence g: $\alpha=\min\{1,\frac{\pi^*/g^*}{\pi^c/g^c}\}$ • $\log\tau^*\sim N(\log\tau^c,s^2)$: extra τ^*/τ^c • Gibbs: $\alpha=1$.

Say: *Stationary distribution = the target π (normalised); with symmetric q, read π off $\alpha=\min\{1,\pi^*/\pi^c\}$.* **Say:** *Acceptance too high (tiny steps) or too low (stuck, flat trace) both mix slowly; aim ≈ 0.2–0.5.*

Traps: no proposal ratio for asymmetric q (≈2 marks) • forward/reverse swapped • dropping $\tau^{n/2}$ • rounding mid-ledger.

13 • Monte Carlo, inversion, rejection

MC: $\mathbb{E}g(\theta)\approx\frac{1}{N}\sum g(\theta^{(s)})$; $P(A)\approx$ mean of indicators; SE $=\text{sd}(g)/\sqrt{N}$ or $\sqrt{\hat p(1-\hat p)/N}$ (independent draws; not the posterior sd). Inversion: $F^{-1}(U)\sim F$; solve $u=F(x)$; Exp(rate λ): $x=-\log(1-u)/\lambda$. Discrete: `x<-0; while(F(x)<U) x<-x+1`.

Rejection (target $\tilde f\propto f$, $Z=\int\tilde f$; proposal g): $X\sim g$, $U\sim U(0,1)$; accept if $U\leq\frac{\tilde f(X)}{Mg(X)}$ [slides: $Y\sim U(0,kh(X))$, accept $Y<f(X)$]. Smallest $M=\sup\tilde f/g$: set $\frac{d}{dx}\log\frac{\tilde f}{g}=0$, check max/endpoints. $P(\text{acc})=Z/M$ ($1/M$ if normalised); tries ~Geom, mean M/Z. Uniform on $[a,b]$: $M=(b-a)\max\tilde f$. Gamma(m,λ) via Exp(μ): best $\mu=\lambda/m$, $k^*=\frac{m^me^{-(m-1)}}{\Gamma(m)}$ ($m>1$). Prior as proposal: accept if $U\leq p(x|\theta)/p(x|\hat\theta_{\text{MLE}})$.

Say: *Rejection needs only $\tilde f$ and gives independent exact draws, but needs an envelope; in high dimensions M is huge and nearly all draws are rejected ⇒ MCMC.*

14 • MCMC diagnostics

Trace: converged = "hairy caterpillar", no trend, constant spread; chains from *overdispersed* starts overlap. Burn-in = discard iterations before chains overlap and stabilise. $\hat R=B/W$: B=sd of pooled post-burn-in draws (all chains), W=mean within-chain sd; $\hat R<1.05$ accept; $\hat R\gg1$ not converged (→ 1 as $T\to\infty$): run longer / better proposal. Lag-k autocorrelation ρ_k (ACF). ESS= $\frac{n}{1+2\sum_{i\geq1}\rho_i}\approx n\frac{1-\rho_1}{1+\rho_1}$; MCMC SE $=s/\sqrt{\text{ESS}}$ (not $s/\sqrt{n}$); aim ESS> 4000. Thinning (every kth): saves memory, loses information, never raises ESS.

Say: *Burn-in from trace: discard up to the iteration where the chains from different starts merge and stop trending (e.g. first _ of _); use the rest.* **Say:** *Run several chains from widely separated starts. Informal: trace plots overlap, stationary, no trend; ACF decays fast. Formal: BGR $\hat R=B/W<1.05$ on post-burn-in draws. Passing doesn't prove convergence: all chains may miss a region.*

15 • Variational inference (CAVI)

$q^*=\arg\min_{q\in\mathcal{D}}\text{KL}(q\|p(\theta|x))$, KL$=E_q[\log q-\log p(\theta|x)]\geq0$ (not symmetric). ELBO$(q)=E_q[\log p(\theta,x)]-E_q[\log q]$; $\log p(x)=$ ELBO + KL, so ELBO$\leq\log p(x)$; max ELBO $\iff$ min KL, no $p(x)$ needed. Mean field $q=\prod_jq_j(\theta_j)$ (assumption on the approximation, not the model).

CAVI: $q_j^*(\theta_j)\propto\exp\{E_{-j}[\log p(\theta,x)]\}$; ELBO never decreases; local optimum; stop when ΔELBO small; several starts, keep highest ELBO. Recipe: log joint line by line, keep terms with θ_j, take E_{-j}, expand, name. $E_\mu[(c-\mu)^2]=(c-E\mu)^2+V\mu$; $E\tau=a^*/b^*$.

Joint prior $\mu|\tau\sim N(m_0,\frac{1}{\kappa_0\tau})$, $\tau\sim$Ga(a,b) (lecture):

$$q_\mu^*=N\left(\frac{\kappa_0m_0+n\bar x}{\kappa_0+n},\ \frac{1}{(\kappa_0+n)E\tau}\right)$$
$$q_\tau^*=\text{Ga}(a^*,b^*),\qquad a^*=a+\tfrac{n+1}{2}$$
$$b^*=b+\tfrac{1}{2}\left[\textstyle\sum(x_i-E\mu)^2+nV\mu\right]+\tfrac{\kappa_0}{2}\left[(E\mu-m_0)^2+V\mu\right]$$

Independent $\mu\sim N(m_0,\frac{1}{\kappa_0})$: $q_\mu^*=N(\frac{\kappa_0m_0+E\tau\,n\bar x}{\kappa_0+nE\tau},\frac{1}{\kappa_0+nE\tau})$, $q_\tau^*=$Ga$(a+\frac{n}{2},\ b+\frac{1}{2}[\sum(x_i-E\mu)^2+nV\mu])$. Iterate $q_\mu\to q_\tau\to\cdots$. IG prior on σ^2: $q_{\sigma^2}^*=$ IG with the same b^*; uses $E[1/\sigma^2]=a^*/b^*$.

Say: *VI vs MCMC: VI optimises: fast, deterministic, scales to big data/many models, but mean-field ignores posterior dependence, underestimates variance, no exactness guarantee. MCMC: slower, needs convergence checks, asymptotically exact.*

19 • Worked mini-examples (fresh numbers, checked in R)

CI for p: logit fit, $\hat\eta_0=0.80$, se= 0.25: $0.80\pm1.96(0.25)=(0.31,1.29)$; $\hat p=\frac{1}{1+e^{-0.8}}=0.69$, CI $(\frac{1}{1+e^{-0.31}},\frac{1}{1+e^{-1.29}})=(0.58,0.78)$: asymmetric.

Quasi F: $D_A=60.0$ (40 df), $D_B=48.0$ (38 df), $\hat\phi_B=1.5$: $F=\frac{12/2}{1.5}=4.00$ on (2, 38), $F_{.95}=3.24$, p= 0.03: keep the term.

E-step: $\pi^0=0.4$, $N(0,1)$ vs $N(3,1)$, $x=1$: $\gamma_1=\frac{0.4e^{-0.5}}{0.4e^{-0.5}+0.6e^{-2}}=\frac{0.2426}{0.2426+0.0812}=0.75$.

MH ledger: $y_1..y_4$ Poisson(λ), $\sum y=10$, $\lambda\sim$Ga$(2,1)\Rightarrow\pi(\lambda)\propto\lambda^{11}e^{-5\lambda}$; $\lambda^c=2$, $\lambda^*\sim$Exp(rate $1/\lambda^c$), $\lambda^*=3$. Target $(\frac{3}{2})^{11}e^{-5(3-2)}=86.4976\times0.006738=0.5828$; proposal $\frac{q(2|3)}{q(3|2)}=\frac{\frac{1}{3}e^{-2/3}}{\frac{1}{2}e^{-3/2}}=1.5340$; $\alpha=\min\{1,0.8940\}=0.89$.